Paper 19 - Observer Face-Selection

CQE-paper-19

CQECMPLX Formal Paper Series

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Construction Basis

Model observation as face selection with latent alternatives retained as obligations, not deleted.

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-19.md
- paper kernel manifest: production\paper-kernels\papers\CQE-paper-19\paper_kernel_manifest.json
- body: production\papers\CQE-paper-19\PAPER-BODY.md
- source: production\papers\CQE-paper-19\SOURCE.md
- formal: production\papers\CQE-paper-19\01-CQE-formal\FORMAL.md
- tool: production\papers\CQE-paper-19\02-CQE-tool\TOOL.md
- workbook: production\papers\CQE-paper-19\03-CQE-workbook\WORKBOOK.md
- recovered_output: production\lib-forge\recovered\papers_output\CQE-paper-19.md

Paper 19 - Observer Face-Selection

Abstract

This paper defines observation as selecting one face of a local registered state while retaining the unselected faces as obligations. The closed result is finite: four frame faces are available, selecting one retains three latent faces, the gluon coordinate `C` is invariant under `L <-> R` antipodal reversal over all eight chart states, and the static `Z4` face template splits as two fixed points plus six period-4 states. Bounded Monster-D4 route evidence supports the observer-face reading after all eight chart states activate, but it remains labeled `pass_with_open_gaps`.

This paper does not prove consciousness, physical measurement collapse, a completed SPINOR signature, or a global observer theorem.

Claims

Claim 19.1. The observer has four selectable frame faces: `C-centroid`, `R-centroid`, `C-flipped`, and `L-centroid`.

Claim 19.2. Selecting one face retains exactly three latent faces.

Claim 19.3. The gluon coordinate `C` is invariant under `L <-> R` antipodal reversal for all eight chart states.

Claim 19.4. The static `Z4` face template has two fixed points, zero period-2 states, and six period-4 states.

Claim 19.5. The bounded observer-route harness provides evidence after all eight chart states activate, but remains open-gap evidence.

Definitions

A **face** is one selectable reading of the local chart: a frame, chirality, or dyad side.

Face selection is the act of committing one face as active.

A **latent face** is an unselected face carried forward with a recovery rule.

The **gluon** is the coordinate `C`, the locally invariant midpoint of the `L | C | R` window.

An **open observer promotion** is any claim that turns this finite selection machinery into consciousness, physical collapse, or an unverified SPINOR signature.

Theorem 19

Observation in the CQE paper suite is admissible as finite face selection with retained latent alternatives:

```
select one face -> keep three latent faces -> preserve C -> record residue
```

and no stronger observer claim is closed here.

Proof

The verifier defines four frame faces. This proves Claim 19.1.

For each face index `0..3`, the verifier selects one face and records the other three as latent.

This proves Claim 19.2.

The gluon-invariance verifier checks all eight chart states. For each state, `gluon(s) = C` and `gluon(swap_LR(s)) = C`. This proves Claim 19.3 and gives the reconstruction rule for antipodal unselected faces.

The `Z4` template verifier reports two fixed points, no period-2 states, and six period-4 states. This proves Claim 19.4.

The Monster-D4 lift harness reports `pass_with_open_gaps` with all eight chart states enumerated and D4 lift preserved after activation. Because the harness also carries open-gap status, it supports the observer-route reading without closing a global observer theorem. This proves Claim 19.5.

Together these claims prove the theorem.

Receipt

Promoted verifier:

```
production/formal-papers/CQE-paper-19/verify_observer_face_selection.py
```

Receipt:

```
production/formal-papers/CQE-paper-19/observer_face_selection_receipt.json
```

Closed layers:

```
four selectable frame faces
one selected face retains three latent faces
gluon C invariant under LR antipodal reversal
static Z4 face template: 2 fixed, 0 period-2, 6 period-4
bounded observer-route evidence after eight-state activation
```

Open layers:

```
SPINOR signature observation
full frame-inversion Q(S) executable binding in the promoted layer
consciousness or measurement-collapse interpretation
global physical observer theorem
```

Falsifiers

The paper fails if selected faces delete latent faces.

It fails if `C` changes under antipodal reversal.

It fails if the `Z4` template contains period-2 states.

It fails if open-gap observer evidence is promoted as a completed theorem.

It fails if SPINOR is claimed observed without a receipt.

Role in the Suite

Paper 18 exports bounded representation routes. Paper 19 says how an observer selects a face of such a route without deleting the alternatives. Paper 20 can therefore ledger each selected face and each retained obligation separately.

Conclusion

Paper 19 closes finite observer face-selection. The selection mechanism is useful precisely because it preserves the unselected faces instead of turning them into hidden assumptions.